Do Center-Balanced Prototypes Generalize? A Controlled Study in Medical Imaging

Anonymous CVPR submission

Paper ID

Abstract

Center balancing and multiple class prototypes are simple choices for transferring frozen visual representations across institutions. We study whether they improve recognition when features, partitions, and source validation are controlled. Our study implementation, center-balanced multi-prototype classification (CBMP), combines inverse group-size weighting, class-conditional center balancing, and spherical clustering. Eight controls separate these choices from discriminative linear and cosine fitting. This design isolates classifier choices within each encoder configuration. We evaluate reviewed PolypGen sequences with ResNet18 and DINOv2 and a custom CAMELYON17-WILDS cohort with ResNet18, UNI, and CONCH. Each comparison uses ten fitting seeds, historical grids, and a four-candidate sensitivity protocol. Under four-candidate selection, primary-hospital trained cosine reaches 97.96% balanced accuracy with UNI and 96.88% with CONCH, compared with CBMP scores of 95.80% and 92.63%. Source validation selects exactly one center-balanced prototype per class for CONCH in all ten seeds, whereas UNI selects four prototypes. Stronger histology representations therefore do not establish a consistent advantage for prototype balancing or multiplicity. Group-pair audits distinguish within-group discrimination from across-group score comparability; patient-macro summaries expose dependence on evaluation weighting. The findings remain conditional on observed cohorts. Full official benchmark evaluation, auditing of pretraining overlap, and independent patient-linked confirmation are required before broader institutional robustness claims.

Keywords: *Medical image classification; domain generalization; prototype learning; cross-center evaluation.*

1. Introduction

Medical image recognition must cope with changes in scanners, acquisition protocols, staining, and patient populations between institutions. A classifier trained at one set of hospitals can lose accuracy on images collected at another hospital, even when the diagnostic labels are unchanged. Cross-hospital pneumonia detection illustrates both this loss and hospital-associated confounding [22]. Recent medical domain-generalization literature treats this as a problem spanning data, representations, models, and evaluation [21]. The relevant question is therefore how to recognize useful class structure without relying on correlations specific to the training institutions.

Pretrained encoders let us compare decision rules while holding the image representation fixed. The value of medical transfer learning depends on the task [16]. Pathology-specific foundation models further motivate separating improvements associated with the encoder from those associated with the classifier [3, 14]. We compare heads on shared features within each encoder configuration: ResNet18 and DINOv2 for endoscopy, and ResNet18, UNI, and CONCH for histology. This tests representation dependence while retaining the distinction between tasks.

Prototype classifiers represent classes with reference vectors. Class means, cosine classifiers, and multiple-prototype models offer different ways to summarize the same features [4, 5, 18]. Multiple references can retain distinct appearances that a single class centroid compresses. Weighting training centers equally changes which observations influence those references. Neither change alone distinguishes diagnostic variation from acquisition effects: a small center can supply a noisy class estimate, and an additional prototype can match a source-specific pattern. Medical prototype methods already address transfer through mechanisms such as dynamic cross-domain prototypes and instance-conditioned classifiers [6, 9]. Their existence makes a controlled comparison with simpler classifiers necessary.

Central question. Do additional prototypes and center balancing improve classification at a held-out institution once features and source validation are fixed? We implement CBMP as a controlled combination of weight-

ing and spherical clustering. We examine both the rule chosen by source validation and fixed settings that isolate its components. A pooled AUROC mixes class separation within groups with score comparisons between groups. Decomposing those pairs can reveal where the performance changes, while inspecting the selected settings establishes which prototype mechanism produced them.

The experiments cover frame-level polyp-presence classification on a reviewed PolypGen subset and tumor-patch classification on a patient-capped CAMELYON17-WILDS subset [1, 2, 12]. PolypGen inclusion requires reviewable center assignments, annotations, and sequence groups. The histology cap is label-blind. The scope is a comparison of frozen-feature heads on these cohorts. Within that scope, the study has three objectives:

1. **Compare decision rules.** Evaluate our CBMP implementation against eight linear, cosine, and prototype controls using the same frozen features and target partitions.
2. **Diagnose the observed differences.** Link the selected settings to fixed prototype-count and center-weighting comparisons, then locate ranking changes within or across recorded groups.
3. **Check dependence on representation and task.** Compare ResNet18 and DINOv2 on endoscopy and ResNet18, UNI, and CONCH on the same histology cohort, keeping features fixed within each head comparison and reporting representation-dependent effects and data limitations.

The contribution is an empirical analysis of classifier construction, source selection, and the groups supporting the evaluation. The ranking audit uses target labels after fitting and is exploratory. Its paired contrasts describe the observed groups with fitted heads fixed.

Section 2 reviews related work; Section 3 defines the classifiers and selection procedure; Section 4 presents the experimental design and findings; and Section 5 discusses their implications and next validation steps.

2. Background and Related Work

2.1. Medical Domain Generalization

Domain generalization evaluates a source-trained model on a target unavailable during fitting [7]. Medical acquisition and population differences can affect appearance and label frequencies [21, 22]. Yoon et al. organize medical DG at the data, feature, model, and analysis levels [21], motivating joint attention to classifier construction and evaluation support.

DomainBed shows why implementation and source-only model selection matter when comparing DG algorithms with empirical risk minimization [7]. Nested evaluation separates tuning observations from those used to estimate performance [19]. In domain adaptation, where unlabeled target images are available, domain invariance and low source error can coexist with poor target performance under certain shifts [23].

2.2. Foundation Encoders and Prototype Classifiers

DINOv2 and DINOv3 develop transferable visual representations through self-supervised learning [15, 17]. In pathology, UNI provides a general-purpose visual representation, while CONCH combines visual and language pretraining [3, 14]. These models motivate asking whether a prototype effect observed with a general-purpose ImageNet encoder persists with pathology-specific representations. We therefore evaluate the original UNI and CONCH image encoders alongside ResNet18 on identical sampled histology patients. The comparison fits the same downstream head families using each model's prescribed image preprocessing; it does not substitute text-prompt classification or reproduce the providers' full evaluation suites. DINOv3 remains related work rather than an evaluated encoder.

Prototypical Networks use class references in a learned metric space [18], and strong cosine-classifier baselines support controlled downstream comparisons [4]. Spherical k-means clusters normalized feature directions [5]. These are the established components of our finite-bank implementation. Temperature scaling adjusts probabilities while preserving the binary score ordering [8]. Consequently, a source-selected temperature alone cannot repair poor across-group ranking.

2.3. Prototype Methods for Medical Transfer

Dynamic prototype-based contrastive learning uses labeled source and unlabeled target data for medical segmentation [6]. InfProto instead trains on a single source and combines learned spatial features with instance-conditioned prototype distributions for segmentation [9]. Its inference can incorporate target-instance and mini-batch statistics. These are distinct access and training settings. Comparing them with our frozen multi-source image heads requires a disclosed task adaptation and optimization budget; the present controls do not establish their relative performance.

PolypGen exposes institutional variation in endoscopy [1]; CAMELYON17-WILDS does so in histology patches [2, 12]. We use them to test a finite weighted bank against discriminative heads and to examine how recorded groups support the apparent transfer.

3. Methodology

3.1. Research Question and Implementation

We test whether center balancing and multiple class prototypes improve transfer beyond learned linear and cosine

heads. To make the comparison explicit, we implement *center-balanced multi-prototype classification* (CBMP) for this study. It combines group/center weighting with weighted spherical k-means [5]. The weighting is a design choice under evaluation; the clustering algorithm and cosine decision rule build on established methods [4, 5].

Training samples are (x_i, y_i, s_i, g_i): image, class, source center, and recorded group. The target institution is excluded from fitting and selection. Figure 1 shows the source-only construction and fixed target path. Objective 1 compares heads on shared features; Objective 2 varies the prototype choices; Objective 3 repeats the comparison across encoder packages and tasks. The concrete datasets, controls, and experimental families are defined together in Section 4.1.

3.2. Frozen Features and Group/Center Weights

Following normalized-feature cosine classification [4], the embedding is

$$z_i = \frac{f_\theta(x_i)}{\|f_\theta(x_i)\|_2}. \tag{1}$$

We use each checkpoint's fixed preprocessing in evaluation mode, without stochastic augmentation. ResNet18 ImageNet1K V1 supplies 512-dimensional pooled features; DINOv2-base supplies the final layer-normalized 768-dimensional CLS embedding [10, 15]. Features must be finite with nonzero norms. All heads in each encoder/task comparison share the same frozen cache. The accompanying provenance pins the model revision, preprocessing, checkpoint, manifest, and feature hashes.

Histology additionally uses original UNI (2024), with 1,024-dimensional CLS features at 224 pixels, and original CONCH, with 512-dimensional unprojected visual features at 448 pixels [3, 14]. Each frozen encoder uses its prescribed transform and the shared cache normalization. Ordered patch identities, heads, candidate grids, source folds, and fitting seeds match the ResNet18 reference. Cross-encoder differences are descriptive: architecture, pretraining, preprocessing and runtime change together. Full extraction metadata is supplied in the supplement.

The ordinary class mean weights each center's class mean in proportion to its sample count [18]. Equalizing centers changes that influence and increases the relative weight of estimates supported by fewer groups.

For class c, let $\mathcal{G}_{s,c}$ contain the groups with class-c training images at center s, let $G_{s,c} = |\mathcal{G}_{s,c}|$, and let $m_{g,c}$ count those images in group g. Our study implementation uses

$$\begin{aligned} a_i &= G_{s_i,c}^{-\beta}/m_{g_i,c}, \\ w_i &= a_i \Big/ \sum_{j:y_j=c} a_j, \qquad \beta \in [0,1]. \end{aligned} \tag{2}$$

Each group has equal total weight within its center and class; a center's unnormalized class weight is $G_{s,c}^{1-\beta}$. Thus four groups outweigh one by four to one at $\beta = 0$, whereas the centers have equal total influence at $\beta = 1$. A longer sequence gains no weight merely through more same-class frames. Available centers can differ by class: balancing supplies no missing appearances and enforces no domain-invariance constraint. Groups are PolypGen sequences and CAMELYON patients; weighting does not establish independence.

3.3. Weighted Spherical Prototype Construction

We extend the spherical clustering objective [5] with the class-normalized weights in Eq. 2. For $\mathcal{I}_c = \{i : y_i = c\}$,

$$\begin{aligned} J_c &= \sum_{i \in \mathcal{I}_c} w_i \min_{1 \le k \le K_c} (1 - z_i^\top p_{c,k}), \\ \|p_{c,k}\|_2 &= 1. \end{aligned} \tag{3}$$

Assignments maximize cosine similarity, resolving ties by prototype index. For assigned set $\mathcal{A}_{c,k}$, the spherical mean update is

$$\begin{aligned} v_{c,k} &= \sum_{i \in \mathcal{A}_{c,k}} w_i z_i, \\ p_{c,k} &= v_{c,k}/\|v_{c,k}\|_2. \end{aligned} \tag{4}$$

This minimizes the fixed-assignment objective when $v_{c,k} \neq 0$ [5]; it does not imply global optimality or robustness.

Initialization samples distinct training rows with probabilities proportional to w_i. Five restarts per class retain the lowest training objective. Empty or near-zero updates are reseeded using the largest weighted residual; exact numerical rules accompany the implementation. Clustering stops at unchanged assignments or 100 updates. K_c is bounded by K, class sample count, and the number of groups containing that class.

3.4. Prediction, Calibration, and Source Selection

The fixed finite bank uses the strongest prototype match for each class:

$$\begin{aligned} h_c(x) &= \max_{1 \le k \le K_c} z(x)^\top p_{c,k}, \\ \hat{y}(x) &= \operatorname*{argmax}_c h_c(x). \end{aligned} \tag{5}$$

This applies a cosine comparison to a spherical cluster bank [4, 5]. At $K = 1, \beta = 0$ it reduces to a group-weighted normalized class centroid; $\beta = 1$ equalizes centers before normalization. Additional prototypes also create more opportunities for accidental matches, which the unbalanced multi-prototype control tests.

Following cosine scoring and temperature scaling [4, 8], probabilities are

$$q(y = c \mid x) = \frac{\exp(h_c(x)/\tau)}{\sum_j \exp(h_j(x)/\tau)}, \quad \tau > 0. \tag{6}$$

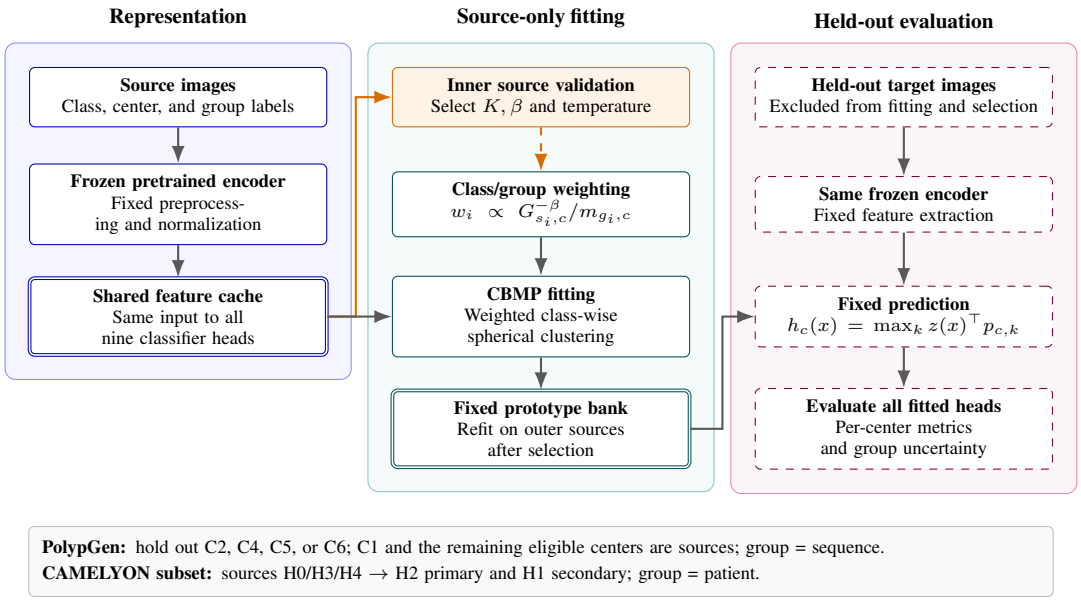


Figure 1. CBMP fitting and evaluation, grouped by function. Comparator heads share the feature cache and partitions; the lower strip specifies the two data protocols. Target evaluation provides no feedback to fitting.

We optimize mean source-validation negative log-likelihood (NLL) over log-temperature bounds $[-4, 4]$. In the binary case, the score margin $h_1(x) - h_0(x)$ determines both the default decision and the ordering of positive-class probabilities. Temperature changes the probability scale while preserving the default decision and ordering. The threshold diagnostic instead moves the decision boundary on that same margin. At inference, the fixed encoder and head score each image independently of its center and other test images.

Inner leave-one-source-center-out validation selects settings by mean balanced accuracy across valid inner centers. Training and validation must be group-disjoint and contain both classes. The supplied outer folds all support center-level validation; the implementation also records a group-disjoint fallback for unsupported cases. Temperature uses the winning configuration's held-out source predictions, after which the head is refitted on all outer sources. Target scores select no setting, checkpoint, or restart [7, 19].

For example, the outer C5 fold uses C1/C2/C4/C6 as sources and rotates inner validation among class-complete sources. C1 contributes negatives to fitting. The selected head is refitted on all four sources. Split checks enforce disjoint sequence IDs and exact image-content hashes.

4. Experiments and Results

4.1. Cohorts, Comparisons, and Uncertainty

The reviewed subset contains 5,444 frames from 37 sequences [1]. Review denotes an annotation/provenance filter, not independent patient-level adjudication. A mask is positive when any pixel exceeds 127; matching bounding-box annotations must agree on presence. Unannotated frames receive negative labels only in documented negative-only sequences. Disputed center assignments and pending annotation review are excluded before fitting. This is not complete-release coverage.

Both classes occur at C2/C4/C5/C6, with 2/10/4/18 sequences, respectively. C1 has negative-only support and remains source-only; C3 is excluded after mapping review. The two encoders share the manifest, partitions, and ten fitting seeds, listed in the supplement. Architecture, pretraining, and preprocessing change together when switching encoders.

The custom CAMELYON17-WILDS cohort contains 17,200 patches from 43 recorded patients, using the existing label-blind cap of 400 patches per patient and original partition [2, 12]. Labels indicate tumor in the central 32×32 region of a 96×96 patch. H0/H3/H4 provide 26 source patients; H2 is the primary target with 9 patients and H1 the secondary target with 8. The same cohort is encoded with ResNet18, UNI, and CONCH. Both targets remain excluded from head fitting, hyperparameter selection, temperature selection, and operating-threshold selection. Standard WILDS instead uses H1 for validation and H2 for testing; our capped cohort and source-only selection define a custom representation-transfer experiment, not a full official benchmark or leaderboard result [20]. The foundation-encoder extension uses previously inspected hospitals and is exploratory rather than independent confirmation.

Prediction unit. Head selection uses frame/patch BA. An additional sequence readout labels presence within reviewed frames by any positive frame, scores a sequence by its maximum raw margin, and fixes its threshold by mean sequence BA at class-complete source-validation centers. C2 has two positive-bearing sequences, leaving binary sequence BA/AUROC undefined. In histology, we average within-patient patch metrics equally across patients. All target patients have positive patches; these summaries do not infer clinical patient or slide diagnoses.

Data ethics. This secondary analysis uses existing dataset releases and involves no new participant recruitment or image collection. The PolypGen authors document frame anonymization and site-specific consent and review procedures [1]. The CAMELYON data collection was approved by the Commissie Mensgebonden Onderzoek regio Arnhem–Nijmegen (2016-2761), with informed consent waived [13].

Experimental families and controls. The main experiment compares nine heads. The component analysis uses fixed $(K, \beta) \in \{1, 4\} \times \{0, 1\}$ and tests sensitivity to the decision threshold and source composition. The source-composition and second-encoder follow-ups are exploratory because we had inspected the original PolypGen results.

The three centroids test which unit contributes equally within a class: image, group, or center. Unbalanced multi-prototype clustering retains inverse group-size weights and fixes $\beta = 0$, so its comparison with CBMP tests whether tuning center balance adds value. The linear and cosine heads learn to separate classes jointly, whereas each prototype bank fits one class at a time. Their weighted variants test group and center weighting in a discriminative objective.

CBMP's original grid has $K \in \{1, 2, 4, 8\}$ and $\beta \in \{0, 0.5, 1\}$. Original candidate counts are 12/4/5/3/15/9 for CBMP, unbalanced prototypes, linear, cosine, weighted linear, and weighted cosine, with one per centroid. A second protocol uses four distinct configurations per tunable family, retaining fixed centroids. CBMP uses $\{1, 4\} \times \{0, 1\}$; the remaining grids are in the supplement. This changes parameter coverage as well as candidate count. Identical source fits are shared across protocols. Each PolypGen encoder supplies 880 target/head/seed evaluations. Each of the three histology encoders supplies 440, giving 3,080 evaluations across five task–encoder configurations. All use ten fitting seeds. These are repeated scores on fixed cohorts, not independent test samples or additional hospitals. The pathology extension adds representations to the existing cohort; it does not expand the patient population. The expanded grids and diagnostics remain exploratory after inspection of the original targets.

Metrics and uncertainty. Balanced accuracy (BA) is the mean of sensitivity and specificity. It gives the two classes equal weight even when a center contributes many more frames of one class. AUROC measures ranking across decision thresholds. Macro F1 summarizes thresholded predictions by class, while NLL and Brier score evaluate predicted probabilities. For the PolypGen summary, we average seed-specific metrics and then give each eligible target center equal weight. Thus C2, with 500 frames, has the same influence on mean BA as C6, with 2,187. Worst-center BA is the minimum of the four seed-averaged center scores. The supplementary materials provide per-center and per-seed metrics and selected settings.

For a paired comparison, both heads are scored on the same resampled groups in each draw. The 2,000 bootstrap attempts (seed 2027) resample whole sequences within each PolypGen target and whole patients within each histology target. We retain only class-complete draws. All predictions from a sampled group move together, preserving dependence among its recorded frames or patches. Fitted heads and the original patch selection remain fixed. The intervals quantify sensitivity to the sampled groups within the observed centers. Seed variation is reported separately because repeating a fit on the same cohort does not supply additional test patients or hospitals.

The supplementary analysis package contains the saved scores, selected settings, input hashes, and scripts needed to reconstruct the ranking diagnostics and paired contrasts.

Execution and assurance. Cached head fitting uses Python 3.12.3, NumPy 1.26.4, SciPy 1.13.1, and scikit-learn 1.5.1. UNI/CONCH extraction uses a separate Python 3.10.22/PyTorch 2.0.1+cu117 environment on a GTX 1660 Ti (6 GB), with deterministic float32 computation, batch 16 and four host threads. Source-only CPU/GPU feature checks and repeated-batch checks pass. The supplement records tolerances, environment versions, checkpoint/feature hashes and exact commands. Saved predictions and frozen source choices support reconstruction; automated checks do not replace author review.

4.2. PolypGen: Center-Specific Transfer

Table 1 reports ten-seed averages and fitting variation. CBMP reaches 58.05% mean BA with ResNet18 and 59.92% with DINOv2, versus 70.20% for linear logistic and 68.35% for trained cosine. Paired CBMP-minus-comparator differences are -12.15 points (conditional sequence-bootstrap 95% interval $[-15.96, -8.34]$) and -8.43 ($[-11.27, -2.90]$). The displayed learned comparators have the highest observed target means; these con-

ditional intervals do not adjust for selecting their names from the table.

Figure 2 shows the site dependence. CBMP's lowest score occurs at C5 with ResNet18 and C2 with DINOv2. DINOv2-CBMP exceeds trained cosine at C6 (77.70% versus 73.12%), but the center average favors cosine. Average and worst-center criteria also select different heads: ResNet18's group-weighted centroid has the strongest worst-center BA (60.34%); DINOv2's weighted cosine leads that criterion (62.25%). Neither is a population worst-case guarantee.

Component diagnostics. Mean BA at $(K, \beta) = (1, 0), (1, 1), (4, 0), (4, 1)$ is 63.15%, 60.48%, 59.78%, 59.42% for ResNet18 and 64.03%, 61.21%, 64.48%, 60.12% for DINOv2. The balancing effect at $K = 4$ minus its effect at $K = 1$ is 2.31 points for ResNet18 (paired 95% interval $[-0.09, 7.23]$) and -1.54 for DINOv2 ($[-6.57, 4.07]$). Both intervals include zero. These ten-seed results do not resolve a uniform mean-BA interaction between multiplicity and balancing.

Tuning sensitivity. With four configurations per tunable family, CBMP reaches 60.56% mean BA with ResNet18 and 60.25% with DINOv2; linear (70.20%) and cosine (68.22%) remain the leading learned heads. The reduced grids change parameter coverage as well as count. Full paired sensitivity results are in the supplement.

Sequence readout. For the same frame-selected heads, the source-thresholded sequence rule gives mean BA over C4/C5/C6 of 39.25% versus 70.14% for ResNet18-CBMP versus linear, and 67.78% versus 60.07% for DINOv2-CBMP versus cosine. The latter point ordering differs from frame BA, but its paired difference of $+7.71$ points has interval $[-4.81, 19.84]$. Aggregation, labels, and operating points change together. Sparse group support and sequence-length effects limit interpretation; C2 remains undefined at this binary endpoint.

4.3. CAMELYON17-WILDS: Frozen Pathology Encoders

We compare ResNet18, UNI, and CONCH on the same custom cohort, with identical target identities, source folds, head families, and fitting seeds. Table 2 focuses on the four-candidate source-only protocol. Historical-grid results and the full nine-head comparisons are reported separately. All heads are frozen and hashed before either target hospital is scored.

Representation changes and classifier differences. At primary H2, trained cosine reaches 97.96% BA with UNI and 96.88% with CONCH, versus 86.73% with ResNet18. CBMP reaches 95.80%, 92.63%, and 85.54%, respectively. Both pathology encoders increase the displayed point estimates, while learned heads retain higher primary-target BA than source-selected CBMP. Encoder-associated gains therefore do not establish an advantage for prototype balancing or multiplicity. AUROC and all-head comparisons are provided in the supplement.

Paired target uncertainty. The H2 CBMP-minus-linear BA difference is -1.92 points for UNI (conditional patient-bootstrap 95% interval $[-5.76, -0.07]$) and -3.87 for CONCH ($[-10.48, -0.52]$). At secondary H1, CONCH-CBMP reaches 93.79% BA versus linear's 92.88%; its $+0.91$-point interval $[-0.50, 2.27]$ includes zero. These exploratory intervals use 2,000 paired patient-resampling attempts with fitted heads fixed, without adjustment for multiple comparisons. They do not quantify a population of hospitals.

Which prototype rule was selected? Under the four-candidate protocol, source validation selects $(K, \beta) = (1, 1)$ for CONCH in all ten fitting seeds. Its selected CBMP therefore reduces to the center-balanced single-centroid case; the reported score is not evidence for a benefit from multiple prototypes. UNI instead selects $K = 4$ in all ten seeds, with $\beta = 0$ in seven and $\beta = 1$ in three. The effective prototype rule depends on the representation and source-validation evidence. These selected configurations motivate reporting both the chosen rule and fixed (K, β) controls; they do not establish why the selected rule transfers or fails at a particular hospital.

4.4. Where Does Ranking Break Down?

Unless stated otherwise, the ranking and patient-weighting diagnostics below use the original candidate grids. The foundation-encoder comparison in Section 4.3 uses the four-candidate sensitivity protocol. The two sets of point estimates should therefore not be interchanged. Group-specific ranking has a precedent in covariate-adjusted ROC analysis [11]. For each held-out center and fitting seed, we compare the positive-class scores of every positive–negative image pair. We record a win when the positive image scores higher and half a win for a tie. The same-group component asks whether a head separates the classes within sequences or patients containing both labels. The across-group component asks whether scores remain comparable between two distinct groups. If ρ is the share of pairs drawn from the same group, then

$$A_{\text{pooled}} = \rho A_{\text{same}} + (1 - \rho) A_{\text{across}}. \tag{7}$$

A component is undefined when it has no eligible pairs, and such a component contributes zero weight to the pooled

Classifier	ResNet18 mean BA	ResNet18 worst BA	DINOv2 mean BA	DINOv2 worst BA
CBMP	58.05 ± 0.68	42.17	59.92 ± 0.48	46.79
Unbalanced multi-prototype	60.99 ± 0.87	53.14	60.18 ± 0.36	47.85
Linear logistic	70.20 ± 0.00	57.17	61.31 ± 0.00	52.12
Trained cosine	63.42 ± 2.46	57.66	68.35 ± 0.85	61.24
Weighted linear	61.78 ± 0.00	47.17	59.24 ± 0.00	51.30
Weighted cosine	63.47 ± 1.89	45.88	66.75 ± 2.50	62.25
Group-weighted centroid	63.15 ± 0.00	60.34	64.03 ± 0.00	53.87
Center-balanced centroid	60.48 ± 0.00	47.63	61.21 ± 0.00	51.53
Image-weighted centroid	60.41 ± 0.00	58.17	62.32 ± 0.00	51.23

Table 1. PolypGen BA (percent) with the original grids. Mean BA is the mean ± fitting-seed SD of ten equal-center averages over C2/C4/C5/C6; worst BA is the minimum center-specific seed mean. The reviewed cohort contains 5,444 frames and 37 recorded sequences; patient linkage is unresolved.

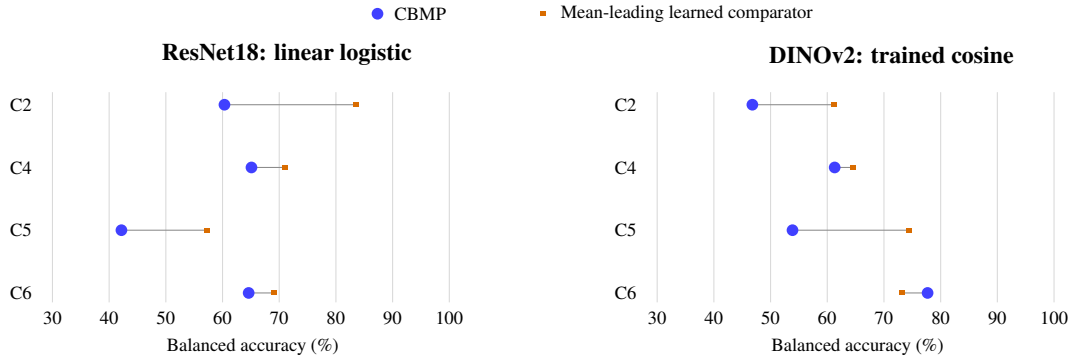


Figure 2. Center-specific PolypGen BA averaged across ten fitting seeds using the original grids. Connectors pair point estimates, not confidence intervals. The displayed learned comparator has the highest observed target mean within its encoder.

Target	Encoder	CBMP	Linear	Cosine
H2	ResNet18	85.54	88.88	86.73
	UNI	95.80	97.72	97.96
	CONCH	92.63	96.50	96.88
H1	ResNet18	77.14	80.30	78.90
	UNI	95.02	97.47	97.48
	CONCH	93.79	92.88	93.95

Table 2. Patch balanced accuracy (%), averaged over ten fitting seeds on identical sampled patients under the four-candidate protocol. H2 is primary and H1 secondary. Linear denotes logistic regression; cosine denotes the trained cosine head. These three heads are a focused view; all nine heads, fitting-seed SDs, AUROC, and the historical-grid controls are reported separately. This is the custom CAMELYON17-WILDS cohort.

score.

Ordinary AUROC implicitly gives a positive group with n_g^+ images and a negative group with n_h^- images weight proportional to $n_g^+ n_h^-$. Equal group-pair AUROC averages observed ordered group pairs uniformly. Class-support inspection identifies removals that make the endpoint undefined. All diagnostics use fixed scores and decisions after fitting. Factorial contrasts use 2,000 sequence-resampling attempts shared across settings, seeds, and encoders. We retain draws with the relevant component defined and report their counts alongside the intervals.

Site/encoder	Head	Pooled	Same	Across	Equal
C5 / ResNet18	CBMP	43.87	93.22	40.43	52.59
	Linear	65.10	91.56	63.26	71.47
C5 / DINOv2	CBMP	57.90	94.86	55.33	64.29
	Cosine	81.95	94.13	81.10	84.52
H2 / ResNet18	CBMP	92.54	91.85	92.59	86.00
	Linear	95.14	94.06	95.21	86.26

Table 3. Ranking diagnostics with the original grids (percent), averaged over ten fitting seeds. Same/Across distinguish recorded group identity; Equal assigns uniform weight to observed group pairs. C5 has one positive-bearing sequence; H2 is the primary histology target.

The selected rule. At C5, all ten DINOv2 fits select $(K, \beta) = (1, 0)$, a group-weighted centroid. An independent calculation through explicit source-group means verifies this analytic limit; its failure there is not unstable multi-prototype optimization. All 199 target positives belong to one sequence, with 51 negatives there and 731 in the other

three. Same-sequence pairs have weight 6.52% in Eq. 7, combining 94.86% Same and 55.33% Across AUROC into 57.90% pooled AUROC. All ten ResNet18 fits instead select $(2, 1)$ at C5, so the observed rules have different mechanisms.

The negative populations also differ: within the mixed sequence they are frames of a polyp-bearing recording, while the others belong to negative-only released sequences [1]. Case mix and between-sequence score shifts are both compatible with the contrast. Removing C5's sole positive-bearing sequence or C2's sole negative-bearing sequence makes frame BA and AUROC undefined. The diagnostic locates a difference without establishing its clinical cause.

Fixed choices and source selection. Holding $K = 4$ fixed at C5, DINOv2 balancing changes Same AUROC from 95.17% to 95.43% and Across from 76.91% to 51.30%. The Across difference is -25.61 points (conditional paired 95% interval $[-45.59, -13.28]$, 1370 valid draws). The ResNet18 difference is -7.55 ($[-21.16, -2.48]$). Table 4 gives all targets; effects vary with site and encoder.

Encoder	Target	ΔSame	ΔAcross
ResNet18	C2	-0.92	-5.20
ResNet18	C4	+7.57	-0.37
ResNet18	C5	+10.84	-7.55
ResNet18	C6	+4.03	-1.34
DINOv2	C2	+7.70	-7.92
DINOv2	C4	+2.11	+1.19
DINOv2	C5	+0.26	-25.61
DINOv2	C6	+1.13	+0.80

Table 4. AUROC change (points) from $\beta = 0$ to $\beta = 1$ at fixed $K = 4$, averaged over ten fitting seeds. Paired conditional intervals and valid-draw counts are supplied in the report and supplement.

Source selection also matters. At C5, DINOv2's $(1, 0)$ setting has 68.00% mean inner source BA, versus 58.00% for $(4, 0)$. Their target BA is 53.87% versus 76.24%. This reversal diagnoses a mismatch between source validation and the observed target. The source-selected head stays fixed throughout the analysis.

Histology and patient weighting. With ResNet18 and the original grids, equal patient-pair weighting at H2 changes the CBMP-minus-linear AUROC gap from -2.59 points to -0.26 with scores unchanged. Patient-macro patch BA is 79.04% versus 80.27%, with paired difference -1.23 ($[-3.98, 2.71]$). At H1 the point ordering reverses: 71.35% versus 70.15%, or $+1.20$ ($[-3.35, 6.75]$). Both macro-BA intervals include zero. These summaries expose dependence on patient weighting; they do not infer clinical patient diagnosis. Full results for both hospitals accompany the report.

4.5. Interpretation and Limits

The algorithmic contrasts locate ranking changes when prototype construction changes, without identifying an acquisition or clinical cause. A uniform group-score shift preserves Same AUROC. Case mix and between-group comparability therefore deserve separate examination rather than attribution to a single nuisance factor.

At C5, resampling reuses one positive-bearing sequence; at C2, Across has one observed group pair. Zero-width conditional intervals can reflect that restricted support. Only 1,026 of 2,000 PolypGen draws retain both frame classes at every target. PolypGen patient linkage and excluded mappings remain unresolved. The earlier three-seed sequence-removal stress analysis is observational and inconclusive. All target-driven diagnostics remain exploratory.

The pathology foundation-encoder comparison addresses representation coverage on the existing cohort, but adds no patients or hospitals. Its conclusions remain conditional on 9 primary and 8 secondary target patients, with no hospital-population inference. Absence of overlap between provider pretraining data and this cohort has not been independently established. The comparison is exploratory because these hospitals had already been inspected. More fitting seeds quantify variability on these same observations; they do not provide independent confirmation. Full official-split evaluation, a modern DG training comparison, resolved PolypGen patient linkage, and an untouched patient-linked cohort remain separate requirements.

5. Conclusion

Across these cohorts, source-selected center balancing and multiple prototypes provide no consistent advantage over learned linear and cosine heads. UNI and CONCH raise histology point estimates, while learned heads retain higher primary-target BA. Source selection also changes the effective mechanism, choosing the single-centroid limit for CONCH and four prototypes for UNI under the four-candidate protocol. Together with the group-pair and aggregate diagnostics, these findings motivate separating representation quality, selected classifier complexity, and prediction unit. Independent patient-linked data, audited pretraining overlap, full official-split evaluation and modern DG controls remain necessary for broader claims.

LLM Usage Statement

OpenAI Codex (GPT-6.1 Sol) assisted with study planning, software implementation and debugging, analysis scripts, and manuscript revision. Google Gemini (3.8

Flash) was used for additional code development. The authors retain responsibility for validating data provenance, implementations, citations, results, and interpretations.

References

[1] Sharib Ali, Debesh Jha, Noha Ghatwary, Stefano Realdon, Renato Cannizzaro, Osama E. Salem, Dominique Lamarque, Christian Daul, Michael A. Riegler, Kim V. Anonsen, Andreas Petlund, Pål Halvorsen, Jens Rittscher, Thomas de Lange, and James E. East. A multi-centre polyp detection and segmentation dataset for generalisability assessment. *Scientific Data*, 10(1):75, 2023. 2, 4, 5, 8

[2] Péter Bandi, Oscar Geessink, Quirine F. Manson, Marcory van Dijk, Maschenka Balkenhol, Meyke Hermsen, Babak Ehteshami Bejnordi, Byungjae Lee, Kyunghyun Paeng, Aoxiao Zhong, Quanzheng Li, Farhad Ghazvinian Zanjani, Svitlana Zinger, Keisuke Fukuta, Daisuke Komura, Vlado Ovtcharov, Shenghua Cheng, Shaoqun Zeng, Jeppe Thagaard, Anders B. Dahl, Huangjing Lin, Hao Chen, Ludwig Jacobsson, Martin Hedlund, Melih Cetin, Eren Halici, Hunter Jackson, Richard Chen, Fabian Both, Jorg Franke, Heidi Kusters-Vandevelde, Willem Vreuls, Peter Bult, Jeroen van der Laak, Bram van Ginneken, and Geert Litjens. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Transactions on Medical Imaging*, 38(2):550–560, 2019. 2, 4

[3] Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, Bowen Chen, Andrew Zhang, Daniel Shao, Muhammad Shaban, Mane Williams, Lukas Oldenburg, Luca L. Weishaupt, Judy J. Wang, Anurag Vaidya, Long Phi Le, Georg Gerber, Sharifa Sahai, Walt Williams, and Faisal Mahmood. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3):850–862, 2024. 1, 2, 3

[4] Wei-Yu Chen, Yen-Cheng Liu, Zsolt Kira, Yu-Chiang Frank Wang, and Jia-Bin Huang. A closer look at few-shot classification. In *International Conference on Learning Representations*, 2019. Preprint version DOI: 10.48550/arXiv.1904.04232. 1, 2, 3

[5] Inderjit S. Dhillon and Dharmendra S. Modha. Concept decompositions for large sparse text data using clustering. *Machine Learning*, 42(1–2):143–175, 2001. 1, 2, 3

[6] Qing En and Yuhong Guo. Unsupervised domain adaptation for medical image segmentation with dynamic prototype-based contrastive learning. In *Proceedings of the fifth Conference on Health, Inference, and Learning*, pages 312–325. PMLR, 2024. 1, 2

[7] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021. Preprint version DOI: 10.48550/arXiv.2007.01434. 2, 4

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017. Preprint version DOI: 10.48550/arXiv.1706.04599. 2, 3

[9] Xiaoqing Guo, Jie Liu, and Yixuan Yuan. Infproto-powered adaptive classifier and agnostic feature learning for single domain generalization in medical images. *International Journal of Computer Vision*, 132(12):5905–5928, 2024. 1, 2

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. 3

[11] Holly Janes and Margaret S. Pepe. Adjusting for covariate effects on classification accuracy using the covariate-adjusted receiver operating characteristic curve. *Biometrika*, 96(2): 371–382, 2009. 6

[12] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton Earnshaw, Imran Haque, Sara M. Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the 38th International Conference on Machine Learning*, pages 5637–5664. PMLR, 2021. The proceedings record does not list a DOI. 2, 4

[13] Geert Litjens, Peter Bandi, Babak Ehteshami Bejnordi, Oscar Geessink, Maschenka Balkenhol, Peter Bult, Altuna Halilovic, Meyke Hermsen, Rob van de Loo, Rob Vogels, Quirine F. Manson, Nikolas Stathonikos, Alexi Baidoshvili, Paul van Diest, Carla Wauters, Marcory van Dijk, and Jeroen van der Laak. 1399 H&E-stained sentinel lymph node sections of breast cancer patients: the CAMELYON dataset. *GigaScience*, 7(6):giy065, 2018. 5

[14] Ming Y. Lu, Bowen Chen, Drew F. K. Williamson, Richard J. Chen, Ivy Liang, Tong Ding, Guillaume Jaume, Igor Odintsov, Long Phi Le, Georg Gerber, Anil V. Parwani, Andrew Zhang, and Faisal Mahmood. A visual-language foundation model for computational pathology. *Nature Medicine*, 30(3):863–874, 2024. 1, 2, 3

[15] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. Preprint version DOI: 10.48550/arXiv.2304.07193. 2, 3

[16] Maithra Raghu, Chiyuan Zhang, Jon Kleinberg, and Samy Bengio. Transfusion: Understanding transfer learning for medical imaging. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2019. 1

[17] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa,

Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. Technical report. 2

[18] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2017. Preprint version DOI: 10.48550/arXiv.1703.05175. 1, 2, 3

[19] Sudhir Varma and Richard Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7:91, 2006. 2, 4

[20] WILDS Benchmark. Leaderboard submissions. `https://wilds.stanford.edu/submit/`, 2026. Accessed November 5, 2026. 4

[21] Jee Seok Yoon, Kwanseok Oh, Yooseung Shin, Maciej A. Mazurowski, and Heung-Il Suk. Domain generalization for medical image analysis: A review. *Proceedings of the IEEE*, 112(10):1583–1609, 2024. 1, 2

[22] John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11):e1002683, 2018. 1, 2

[23] Han Zhao, Remi Tachet des Combes, Kun Zhang, and Geoffrey J. Gordon. On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning*, pages 7523–7532. PMLR, 2019. Preprint version DOI: 10.48550/arXiv.1901.09453. 2